

Assignment 04

a) $f(x) = e^{0.5x} + \sin x$ $[0, 2]$

$$I(f) = \int_0^2 (e^{0.5x} + \sin(x)) dx$$

Antiderivatives

$$\begin{aligned} F(x) &= \frac{e^{0.5x}}{0.5} - \cos(x) \\ &= 2e^{0.5x} - \cos x \end{aligned}$$

Evaluation

$$I(f) = [2e^{0.5(2)} - \cos(2)] - [2e^{0.5(0)} - \cos(0)]$$

$$I(f) = 2e^1 - \cos(2) - 1 = 4.852$$

b) The closed newton-cotes formula for $n=1$ is the Trapezoidal Rule, given as

$$J_1 = \frac{b-a}{2} [f(a) + f(b)]$$

Here, $a=0$ & $b=2$

$$f(0) = e^0 + \sin(0) = 1$$

$$f(2) = e^1 + \sin(2) = 3.627$$

$$J_1(f) = \frac{2-0}{2} [1 + 3.627]$$

$$J_1(f) = 1 \times 4.627 = 4.627$$

c) Simpson's rule ($n=2$) is given by the formula

$$\frac{b-a}{6} [f(a) + 4f(m) + f(b)], \text{ where } m = \frac{a+b}{2}$$

$$\text{Here, } m = \frac{0+2}{2} = 1$$

$$\therefore f(0) = 1$$

$$f(1) = e^{0.5} + \sin(1) = 2.490$$

$$f(2) = 3.627$$

$$\begin{aligned}\therefore J_2(f) &= \frac{2-0}{6} [1 + 4(2.490) + 3.627] \\ &= \frac{1}{3} [1 + 9.960 + 3.627] \\ &= 4.862\end{aligned}$$

d) This refers to applying the trapezoidal rule over multiple sub-intervals. This formula is:

$$C_{1,m} = \frac{1}{2} [f(x_0) + 2f(x_1) + \dots + 2f(x_{m-1}) + f(x_m)]$$

For 4 subdivisions ($m=4$) the step-size is,

$$\begin{aligned}h &= \frac{b-a}{m} \\ &= \frac{2-0}{4} = 0.5\end{aligned}$$

The nodes are,

$$x_0 = 0, x_1 = 0.5, x_2 = 1, x_3 = 1.5, x_4 = 2$$

$$f(0) = 1$$

$$f(0.5) = e^{0.25} + \sin(0.5) = 1.763$$

$$f(1) = 2.490$$

$$f(1.5) = e^{0.75} + \sin(1.5) = 3.114$$

$$f(2) = 3.627$$

$$\begin{aligned}C_{1,4}(f) &= \frac{0.5}{2} [1 + 2(1.763) + 2(2.490) + 2(3.114) \\ &\quad + 3.627] \\ &= 4.84097\end{aligned}$$